Measure pressure drop through an empty packet in wind tunnel

10 messages

Justin Flory <Justin.Flory@asu.edu> Mon, Oct 19, 2026 at 8:00 AM
To: Samantha Taylor <Samantha.Kneale@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>
Cc: Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Hi Samantha and Lindsey,

I'd like to introduce you to Prof. Marcus Herrmann and Leo Dayal. They are building a CFD model of the CO2 capture from a mesh packet for the FLEX-DAC project. They would like to get a pressure drop measurement through a mesh. It coudld be a packet or just a sheet of mesh. Please make sure to not the pore size and percent open. This will likely require a different holder but could be just adhered across the face of the empty holder.

Eventually, we'll also want pressure drop measurements of the packets mounted as they normally are, with flows parallel to the packet, but that will come later for the system design. If its easy to do with the other test then by all means do them both.

Is this something you'd be able to do by Oct. 23?

I'm also attaching pressure drop data collected on Excellion sheets.

Justin

Milestone 2.1.pdf
121K

Samantha Taylor <Samantha.Kneale@asu.edu> Mon, Oct 19, 2026 at 3:34 PM
To: Justin Flory <Justin.Flory@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>
Cc: Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>, "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>

Hi All,

I'm looping in Ramesh here while Lindsey is out of office. He has already conducted wind tunnel pressure drop tests with the MT 125 micron polyester mesh (through two layers as a large packet) and has agreed to send us that data. I've also provided him a 25 micron nylon mesh to test as soon as he is able. I'll let him answer about the possible timeline.

Thanks,

Samantha Kneale Taylor

Laboratory & Research Manager

Center for Negative Carbon Emissions

Arizona State University

Cell: (602)405-4586
Office: (602)496-8516

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Ramesh Babu Komma <rkomma1@asu.edu> Mon, Oct 19, 2026 at 5:45 PM
To: Samantha Taylor <Samantha.Kneale@asu.edu>
Cc: Justin Flory <Justin.Flory@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

All,
Attached are the results I collected in August for the MT 125 micron mesh with a custom 3D printed stand inside the wind tunnel. I can provide the dataset for the 25 micron nylon mesh by 23rd October.
Best,
Ramesh Komma
[Quoted text hidden]

MT 125 micron Mesh dP data RK 10:19.docx
377K

Justin Flory <Justin.Flory@asu.edu> Mon, Oct 26, 2026 at 10:54 AM
To: "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>, Samantha Taylor <Samantha.Kneale@asu.edu>
Cc: Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Thanks, Ramesh. This is quite a bit higher than when running parallel to the sheets, but I think it's what Marcus and Leo need. When do you expect to share the data from the 25-micron nylon mesh?

Can you also please share the percent openness for each of these meshes?

Thanks,
Justin

From: Ramesh Babu Komma <rkomma1@asu.edu>
Sent: Monday, October 19, 2026 5:45 PM
To: Samantha Taylor <Samantha.Kneale@asu.edu>
Cc: Justin Flory <Justin.Flory@asu.edu>; Lindsey Hamblin <lmhambli@asu.edu>; Marcus Herrmann <marcus.herrmann@asu.edu>; Leonardo Dayal (Student) <ldayal@asu.edu>
Subject: Re: Measure pressure drop through an empty packet in wind tunnel

[Quoted text hidden]

Ramesh Babu Komma <rkomma1@asu.edu> Mon, Oct 26, 2026 at 1:04 PM
To: Justin Flory <Justin.Flory@asu.edu>
Cc: Samantha Taylor <Samantha.Kneale@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

All,
I'm encountering high pressure drops (32 Pa at just 0.6 m/s) and significant sensor fluctuations with the 25-micron nylon mesh, which limits the testable wind speed range. I am confident in the earlier dataset and hope it provides sufficient coverage for the study. However, if specific target speeds are still needed, please let me know and I will gladly gather additional data.

Best Regards,
Ramesh
[Quoted text hidden]

Justin Flory <Justin.Flory@asu.edu> Mon, Oct 26, 2026 at 1:27 PM
To: "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>, Samantha Taylor <Samantha.Kneale@asu.edu>
Cc: Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Thanks, Ramesh. Attached is some analysis Samantha did on a few other mesh materials, measuring the percent openness, including the 25-micron (A25 uncoated) mesh, which had 8% open, which may explain why it had such a high pressure drop.

@Samantha can you help measure the percentage open on at least the 125-micron mesh?

Justin

Get Outlook for Android

From: Ramesh Babu Komma <rkomma1@asu.edu>
Sent: Monday, 26 October 2026 13:04:13
To: Justin Flory <Justin.Flory@asu.edu>
Cc: Samantha Taylor <Samantha.Kneale@asu.edu>; Lindsey Hamblin <lmhambli@asu.edu>; Marcus Herrmann <marcus.herrmann@asu.edu>; Leonardo Dayal (Student) <ldayal@asu.edu>
[Quoted text hidden]

[Quoted text hidden]

2 attachments

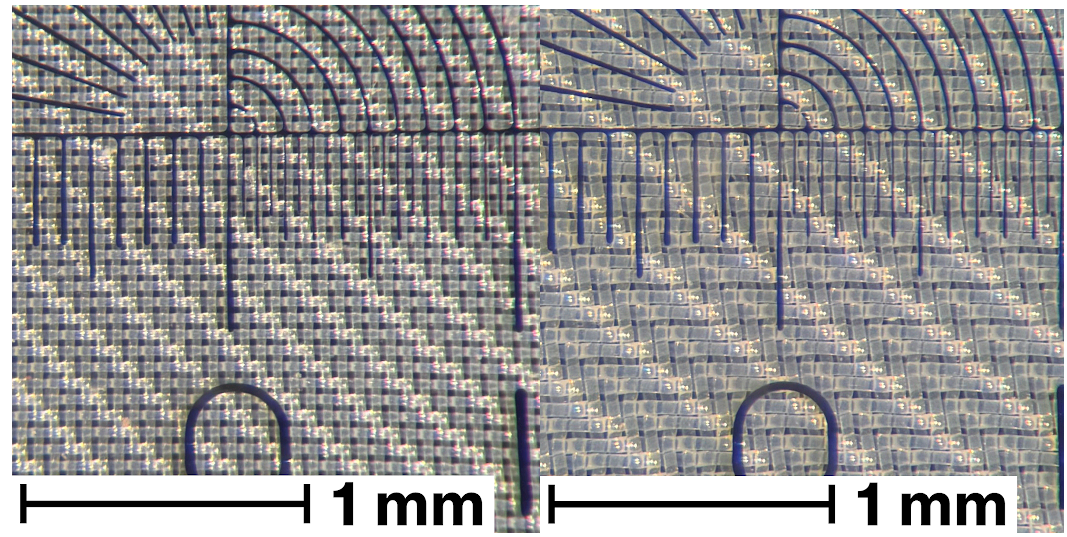
Progress Report_ASU_Vermaas_EE0009274_FY2023-Q3 1.0.pdf
4412K

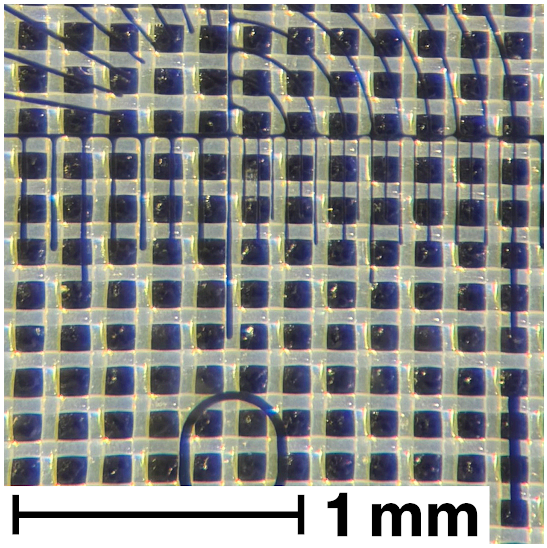
Mesh Openness Analysis - AUDACity report FY2023-Q3.pdf
1298K

Samantha Taylor <Samantha.Kneale@asu.edu> Tue, Oct 27, 2026 at 1:53 PM
To: Justin Flory <Justin.Flory@asu.edu>, "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>
Cc: Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Hi Justin,

The mesh Ramesh used for this WT run was actually the CellMicroSieves 25 micron mesh (used because the weave was more consistent):





Left to right: CMS25, A25, MT125

I assume a slightly higher percent open than the A25, but I will need to do analysis on both CMS25 and the MT125. I will work on that this week.

[Quoted text hidden]

Samantha Taylor <Samantha.Kneale@asu.edu> Mon, Nov 2, 2026 at 5:29 PM
To: Justin Flory <Justin.Flory@asu.edu>, "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>
Cc: Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Hi All,

I created new histograms for percent open on the CellMicroSieves 25 micron (CMS25) and Sefar Mechanical Tree 125 micron (MT125) meshes (see ppt attached for method). I also did a run on the AZPressCo 25 micron mesh, which is the standard we used in the algae trials. I got a lower percent open than the Amazon (A25), even though I suspect they had the same manufacturer. Could be the quality of the image, or a slightly different tolerance level on the selection, but I tried to get them as consistent as possible.

CMS25: 12.02%

MT125: 32.98%

APC25: 4.51%

I have also attached the dimensions of Lindsey's windtunnel jig Ramesh used for the pressure drop tests. We will still need to do the blank and filled packet wind tunnel runs, but I will update as we progress on that.

Thanks,

Samantha Kneale Taylor

Laboratory & Research Manager

Center for Negative Carbon Emissions

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2 attachments

2026 percent open histograms.pptx
6677K

windTunnelHolder.pdf
34K

Ramesh Babu Komma <rkomma1@asu.edu> Thu, Nov 5, 2026 at 2:17 PM
To: Samantha Taylor <Samantha.Kneale@asu.edu>
Cc: Justin Flory <Justin.Flory@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Hi all,

I have conducted a trial with packed beads in the wind tunnel. Attached are the results from all the runs I've completed so far, which include one filter, a blank with a frame, and packed beads in a 40-micron filter.

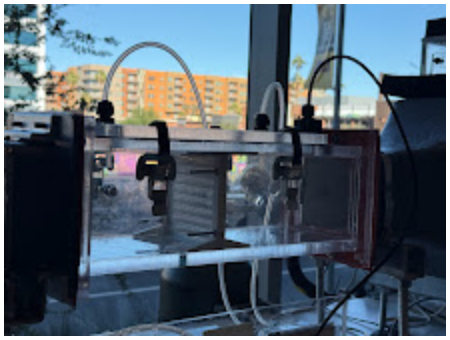
Thank you to Samantha for helping with the package. She will provide further details regarding the packing density soon. Hope this helps, and I am happy to answer any questions.

Best,

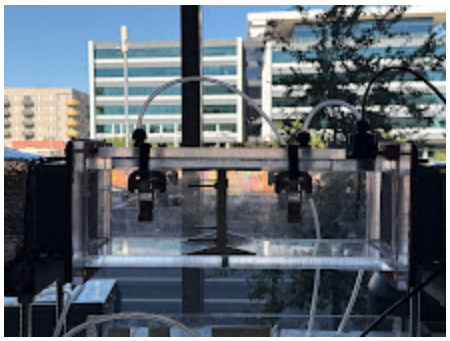
Ramesh Komma

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3 attachments



IMG_8879.HEIC
1244K



IMG_8880.HEIC
1783K

Pressure Drop of Filter.xlsx
12K

Samantha Taylor <Samantha.Kneale@asu.edu> Thu, Nov 5, 2026 at 3:25 PM
To: "Ramesh Babu Komma (Student)" <rkomma1@asu.edu>
Cc: Justin Flory <Justin.Flory@asu.edu>, Lindsey Hamblin <lmhambli@asu.edu>, Marcus Herrmann <marcus.herrmann@asu.edu>, "Leonardo Dayal (Student)" <ldayal@asu.edu>

Hello All,

The packet has a fill space of 10 cm x 10 cm (opening of the frame). There are 8 horizontally stacked, individual sorbent pockets (aka sticks), with a width of 1.25 cm (minus width of thermal seal element). Each stick has a fill of ~0.3 g unexchanged Purolite A501P (chloride form).

The sorbent was stored humid; I dried it for a few hours in the hood before packaging, but even so actual dry sorbent weight may be slightly lower.

Mesh used was CMS25: 25 micron pore size with approx. 12.02 % open, nylon

Thanks,

Samantha Kneale Taylor

Laboratory & Research Manager

Center for Negative Carbon Emissions

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